

Radiotherapy for eye cancer (ocular melanoma)

Radiotherapy uses high-energy rays to destroy eye melanoma cells.

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About radiotherapy for eye cancer

Radiotherapy uses high-energy rays to destroy the cancer cells, while doing as little harm as possible to normal cells. For [eye cancer](#) (ocular melanoma), you may have radiotherapy in either of the following ways:

- You may have a small radioactive disc put on the outside of the eye. This is called [brachytherapy](#). It is the most common way of giving radiotherapy for eye melanoma.
- You may have it from a radiotherapy machine. This is called [external radiotherapy](#). It sometimes uses specialised techniques.

You may have radiotherapy:

- as your main treatment – this is more common for [uveal melanoma](#)
- to shrink the melanoma before [surgery](#)
- after surgery or chemotherapy eye drops, to reduce the risk of the melanoma coming back – this is called adjuvant radiotherapy.

Adjuvant radiotherapy is usually a treatment for [conjunctival melanoma](#).

Brachytherapy

Brachytherapy is the most common way of giving radiotherapy for uveal and conjunctival melanoma. It uses a small radioactive disc called a plaque.

Brachytherapy is used to treat small to medium-sized eye melanomas in an area that can be easily reached to attach the disc.

You have an operation to attach the plaque to the wall of the eye over the tumour. This operation is usually done under general anaesthetic. But it can be done using an injection of local anaesthetic to numb the area.

The nurses will give you drugs to help you relax before the doctor attaches the plaque.

The plaque is left in place until you have had the right amount of radiation. This is usually 1 to 5 days. After this, you have another short operation to remove the plaque.

You stay in hospital for up to 1 week during treatment. While the plaque is attached to your eye, there is a small risk of radiation exposure for the people around you. Your doctor and nurse will explain this. They will tell you about the safety measures that are necessary.

You will need to stay in one room and visitors will only be allowed in for a short time. Children and anyone who is pregnant will not be allowed to visit. This is because of the risk of radiation.

When the plaque is removed, you can usually go home on the same day or the next day.

External radiotherapy

External radiotherapy aims high-energy rays from a machine at the tumour. It destroys cancer cells in the area it is aimed at. It does not make you radioactive.

The main types of external radiotherapy used to treat eye melanoma are:

- proton beam radiotherapy
- stereotactic radiation therapy.

These are specialised techniques that help protect healthy tissue and protect your eyesight. If you have uveal melanoma, you may have external radiotherapy on its own. If you have conjunctival melanoma, you may have external radiotherapy and chemotherapy eye drops.

Proton beam radiotherapy

[Proton beam radiotherapy](#) uses proton radiation to destroy cancer cells. The proton beam is aimed directly at the tumour. This causes very little damage to surrounding healthy tissue.

Proton beam radiotherapy is only available in a few cancer hospitals in the UK. It is used to treat uveal and conjunctival melanoma. This is usually when the melanoma cannot be fully treated with brachytherapy because of its size or position.

You may need to have a [mask or mould](#) made before treatment. This is to help you stay still and in the correct position during your proton beam therapy.

Before treatment, you have an operation to attach tiny metal clips to the white outer wall of the eye. These clips are called tantalum markers. You have this operation under general anaesthetic. You cannot see the clips, but they show up on scans. This helps your doctors plan the treatment. Because the clips are harmless, they are left in place after treatment unless you find them uncomfortable.

You have proton beam radiotherapy in small doses. These doses are called fractions. Each fraction is given over a few minutes each day. You usually have proton beam radiotherapy for 4 to 5 days as an outpatient.

Stereotactic radiotherapy

[Stereotactic radiotherapy](#) uses many small beams of radiation to target the tumour. It delivers high doses of radiotherapy to very precise areas of the eye. This reduces the risk of side effects. You may have it if proton beam radiotherapy is not available.

Before having stereotactic radiotherapy, you have an [MRI scan](#). This helps doctors plan your treatment.

Your doctor gives you an injection of local anaesthetic to stop your eye from moving. You may need to wear a light-weight metal head frame to keep you in the correct position.

The treatment usually takes 45 minutes. You usually only need 1 session of treatment, as a day patient.

Side effects of radiotherapy

The side effects will depend on the type of radiotherapy and where the tumour is in the eye. Some side effects may be caused by the surgery to apply the brachytherapy plaque or tantalum markers to the eye.

Your doctor, nurse or radiographer will explain what to expect during treatment and the possible side effects. They will tell you how side effects are managed and what you can do.

Side effects may last up to a few weeks after treatment. Some side effects may not improve. Or they may develop months or years after treatment. These are called late or long-term side effects.

Immediate side effects

Immediate side effects may include:

- double vision or blurry vision – this usually improves over a few weeks
- a swollen or red eye – you can have eye drops or cream to improve this
- a dry eye – you can use artificial tears or eye drops until this improves
- loss of eyelashes
- increased pressure in the eye due to swelling – you can have eye drops or drugs called steroids to treat this.

Tell your eye doctor or nurse straight away if you have any changes in your eyesight at any time. Sometimes dryness of the eye is a long-term side effect.

Different things may help to improve side effects. Always tell your eye doctor or nurse if you have any side effects.

Long-term or late side effects

Possible long-term or late side effects include:

- cataracts (cloudiness in the lens of your eye) – because of damage to the lens of the eye
-

permanent changes to your eyesight – because of damage to certain areas of the eye

- pain in the eye.

Cataract surgery, laser treatment and injections of a drug called [bevacizumab](#) may be helpful to treat some late effects. If the eye becomes very painful, you and your doctor may talk about whether to [remove the eye](#).

About our information

References



Below is a sample of the sources used in our eye cancer (ocular melanoma) information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Jain P, Finger PT, Fili M, et al. Conjunctival melanoma treatment outcomes in 288 patients: a multicentre international data-sharing study. *British Journal of Ophthalmology* 2021;105:1358–1364. (accessed May 2022).

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, Professor Samra Turajlic, Consultant Medical Oncologist.

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Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help